Dual-Stream Progressive Hazard Network (DSPHN): A Novel Architecture Combining Appearance-Context Disentanglement with Hierarchical Safety-Aware Inference for Indoor Hazard Segmentation

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Abstract

Indoor environments represent one of the most underappreciated sources of serious injury in daily life. Falls, collisions with furniture and clutter, and lacerations from sharp objects collectively account for millions of emergency hospital visits annually, yet no existing computer vision system is designed specifically to detect and localise these hazards at the pixel level from a standard RGB camera. General-purpose semantic segmentation architectures treat hazardous and non-hazardous regions identically, providing no mechanism to exploit the inherent two-level hierarchical structure of hazard understanding: first determining whether a region is dangerous at all, then classifying the specific type of danger present.

*We present the **Dual-Stream Progressive Hazard Network (DSPHN)**, a novel semantic segmentation architecture that principally integrates two previously separate paradigms for the first time. The first paradigm, inspired by the Dual-Attention Hazard Decoder (DAHD), employs parallel Appearance and Context decoder streams with severity-guided Fusion Gates to disentangle what hazards look like from where they occur spatially. The second paradigm, inspired by the Progressive Hazard Refinement Network (PHRN), implements hierarchical safety-aware conditional inference through a two-stage design: Stage 1 produces a binary safe/unsafe prediction map from the fused dual-stream features, and Stage 2 performs fine-grained four-class hazard classification conditioned on the Stage 1 output. Critically, the Stage 1 output is a direct structural input to Stage 2: not an auxiliary loss signal or attention weight: making Stage 2 spatially aware of hazard locations from the very first classification layer. Neither paradigm alone constitutes the architectural contribution; the principled integration of dual-stream appearance-context disentanglement with hierarchical safety-aware conditional inference is what is novel.*

*We further introduce a **four-class indoor hazard taxonomy**: Background, Safe Floor, Obstacle, and Sharp Objects: grounded in real-world injury risk profiles rather than semantic identity, and a **manual hazard annotation dataset** of 100 ADE20K indoor images with pixel-level labels created using Make Sense AI, the first annotation set combining floor safety, obstacle, and sharp object categories in a unified indoor segmentation framework. The training dataset is augmented with 186 COCO knife and scissors images for additional Sharp Objects supervision, applied without floor-level spatial constraint since sharp objects constitute laceration hazards regardless of their vertical position in a scene.*

*A controlled ablation study trains all three variants: DAHD-only, PHRN-only, and the integrated DSPHN: under identical conditions. DSPHN achieves test Mean IoU of **0.4198** and Pixel Accuracy of **0.8332**, outperforming DAHD-only (mIoU 0.3894) by +0.030 and PHRN-only (mIoU 0.3793) by +0.041, confirming that the integrated design is quantitatively superior to either component in isolation. Strong per-class results on Background (IoU 0.8081) and Safe Floor (IoU 0.5517) demonstrate reliable hazard region localisation. Sharp Objects achieves zero IoU due to extreme pixel-level sparsity ($< 0.05\%$ of test pixels), a fundamental long-tail challenge that motivates future targeted data collection.*

1. Introduction

According to the World Health Organization, falls are the second leading cause of accidental injury deaths globally, with the vast majority occurring in indoor domestic environments [19]. The scale of this problem is significant: the United States Centers for Disease Control and Prevention estimates that falls cause over 36,000 deaths and 8 million emergency department visits annually in the US alone. Beyond falls, lacerations from sharp objects–knives, scissors, and broken glass–represent a further substantial source of domestic injury, particularly among children under five and

adults over sixty-five. Furniture and clutter obstacles contribute additional collision and entrapment injuries across home, hospital, and retail environments.

The development of intelligent domestic systems creates an urgent need for vision systems capable of parsing indoor scenes in terms of their *safety consequences* rather than purely their semantic content. Assistive robots navigating hospital corridors, smart monitoring systems for elderly residents, autonomous wheelchairs operating in homes, and occupational safety inspection systems all require pixel-level knowledge of which regions are safe to traverse and which present hazards. A system that cannot distinguish a wet slippery floor from a dry safe floor, or that classifies a knife on the ground identically to a harmless book, is fundamentally inadequate for safety-critical deployment.

Semantic segmentation–the task of assigning a class label to every pixel in an image–provides the natural representational framework for this problem. However, applying existing segmentation architectures to indoor hazard understanding exposes two fundamental gaps that current methods do not address.

Gap 1: Missing hazard-specific annotations. No existing segmentation benchmark defines pixel classes aligned with indoor injury risk. ADE20K [22], the largest indoor scene parsing benchmark, provides 150 semantic categories but treats all floor types identically and has no concept of obstacle severity or laceration risk. The COCO dataset [8] provides instance segmentation for everyday objects including knives and scissors, but does not frame these in terms of safety consequence. NYU Depth V2 [14] and SUN RGB-D [15] similarly provide generic indoor annotations without safety-critical class definitions.

Gap 2: Single-level simultaneous prediction. Every existing semantic segmentation architecture–U-Net [13], DeepLab [3], PSPNet [21], SegFormer [20]–predicts all semantic classes simultaneously from a single decoder pass over encoder features. This design provides no mechanism to first answer whether a region is dangerous before classifying its specific hazard type, missing a natural two-level hierarchy inherent to safety reasoning. A human inspector, when assessing indoor hazard risk, naturally first identifies which regions deserve attention (coarse binary detection) and then classifies the specific hazard type (fine-grained recognition). No existing architecture models this reasoning structure.

This paper addresses both gaps through a unified architectural and data contribution. For Gap 1, we introduce the first hazard-specific segmentation taxonomy and contribute a manually annotated indoor hazard dataset. For Gap 2, we propose DSPHN, which integrates two previously separate architectural paradigms into a single model:

1. **DAHD dual-stream decoding:** Parallel Appearance and Context decoders with severity-guided Fusion Gates disentangle the two key aspects of hazard recognition: local visual appearance cues (*what* does a hazard look like?) and spatial contextual cues (*where* does a hazard occur relative to its surroundings?).
2. **PHRN hierarchical inference:** A two-stage conditional design where Stage 1 first produces a binary safe/unsafe prediction from the fused dual-stream features, and Stage 2 receives this binary map as a structural input to perform fine-grained four-class hazard classification.

The specific contributions of this paper are:

- **DSPHN: Novel Integrated Architecture.** The first segmentation architecture to combine dual-stream appearance-context disentanglement with hierarchical safety-aware conditional inference for indoor hazard segmentation from a single RGB image. The combination produces emergent benefits: Stage 1 maps derived from dual-stream features are more accurate than single-decoder Stage 1 maps, providing better conditioning for Stage 2 and explaining why DSPHN outperforms both parent architectures.
- **Controlled Ablation Validation.** All three variants (DAHD-only, PHRN-only, DSPHN) are trained and evaluated under identical conditions. DSPHN achieves mIoU 0.4198 versus 0.3894 (DAHD-only) and 0.3793 (PHRN-only), demonstrating that integration is quantitatively superior to either component alone.
- **Indoor Hazard Taxonomy.** A four-class safety-oriented taxonomy–Background, Safe Floor, Obstacle, Sharp Objects–defined by real-world injury consequence rather than semantic identity, enabling training signal directly aligned with safety outcomes.
- **Manual Hazard Annotation Dataset.** 100 ADE20K indoor images manually annotated at pixel level using Make Sense AI with three hazard categories, constituting the first annotation set combining floor safety, obstacle, and sharp object categories for indoor segmentation.
- **COCO Sharp Object Augmentation.** Integration of 186 COCO knife and scissors instances without floor-level spatial constraint, motivated by the observation that sharp objects constitute laceration hazards regardless of their vertical position in a scene.

The remainder of this paper is organised as follows. Section 2 reviews related work in semantic segmentation, multi-stage architectures, attention mechanisms, loss functions, and indoor safety understanding. Section 3 describes the DSPHN architecture in full mathematical detail. Section 4 presents the dataset construction and annotation methodology. Section 5 reports experimental results including ablation study, per-class analysis, training dynamics, qualitative evaluation, and extended discussion. Section 6 concludes with future directions.

2. Related Work

2.1. Semantic Segmentation Architectures

The foundational work of Long *et al*. [10] established fully convolutional networks (FCNs) for dense pixel prediction, replacing classification heads with upsampling paths to enable end-to-end training for per-pixel label assignment. The encoder-decoder paradigm was significantly refined by Ronneberger *et al*. [13], whose U-Net architecture introduced symmetric skip connections between encoder and decoder stages to preserve high-frequency spatial details lost during downsampling, originally demonstrated on biomedical image segmentation. U-Net's elegant design has since become a backbone architecture across diverse domains including remote sensing, medical imaging, autonomous driving, and indoor scene understanding.

DeepLab [3] addressed the need for multi-scale context through atrous (dilated) convolution and Atrous Spatial Pyramid Pooling (ASPP), enabling the network to capture information at multiple receptive field scales without increasing the number of parameters. PSPNet [21] employed a Pyramid Pooling Module to aggregate global scene context through multi-resolution pooling. Subsequent transformer-based architectures have achieved state-of-the-art results on benchmark datasets: SegFormer [20] combines a hierarchical Mix Transformer encoder with a lightweight all-MLP decoder, while Swin-UNet [2] adapts the Swin Transformer for pure-transformer medical image segmentation.

A critical observation underlying DSPHN is that *all* of these architectures share a common design principle: a single decoder that predicts all semantic classes simultaneously from encoded features. This design treats hazard and non-hazard regions identically, providing no hierarchical structure for safety-aware reasoning. DSPHN departs from this paradigm by introducing a two-stage conditional decoder.

2.2. Multi-Stream and Dual-Decoder Architectures

Dual-decoder and multi-stream designs have been primarily explored in RGB-D segmentation, where separate processing branches handle the RGB and depth input modalities [4]. The bi-directional modality fusion approach of Chen *et al*. [4] uses cross-modal attention to propagate features between the two streams. In the single-modality RGB setting, dilated convolutions have been used in DeepLab variants to capture multi-scale context, but within a single decoder pathway.

DSPHN applies parallel Appearance and Context decoders to a *single RGB image*. The dual-stream design is not driven by two input modalities but by the semantic distinction between two aspects of hazard recognition that benefit from different inductive biases: standard convolutions for local texture pattern learning (Appearance Decoder) and dilated convolutions for spatial relationship modelling over larger receptive fields (Context Decoder). This is a qualitatively different motivation from RGB-D dual-stream designs.

2.3. Coarse-to-Fine and Multi-Stage Architectures

Coarse-to-fine prediction strategies have been applied across multiple vision tasks. Eigen *et al*. [5] used a two-stage coarse-to-fine network for depth estimation, where a global prediction is refined at higher resolution. Cascade R-CNN [1] employs sequential bounding box regression stages for object detection, where each stage receives refined proposals from the previous. In semantic segmentation, label refinement networks [7] refine coarse predictions at progressively finer resolutions.

DSPHN's two-stage design is qualitatively different from all of these. Standard coarse-to-fine approaches refine the *same output type* (depth, bounding box, segmentation map) at increasing resolution. In DSPHN, Stage 1 and Stage 2 answer *categorically different questions*: Stage 1 performs binary safe/unsafe detection, while Stage 2 performs four-class hazard classification. Stage 1 output is a different type of prediction entirely, not a coarser version of Stage 2 output.

2.4. Attention Mechanisms in Segmentation

Attention mechanisms have been widely adopted to improve feature selectivity. The Squeeze-and-Excitation Network [6] introduced channel-wise attention via global average pooling to recalibrate channel responses. Non-local networks [18] modelled long-range spatial dependencies through self-attention. Attention U-Net [12] proposed soft spatial attention gates on skip connections, using decoder gating signals to suppress irrelevant encoder activations.

DSPHN's Severity Fusion Gates differ from these in three ways: (1) they blend two *complete decoder pathways* rather than modulating individual skip connections; (2) they are explicitly initialised from domain hazard severity scores providing a principled starting point for fusion learning; and (3) they learn when to trust appearance versus spatial context for each pixel, rather than learning which encoder features to suppress.

2.5. Loss Functions for Class Imbalance

Class imbalance is pervasive in semantic segmentation. Class-weighted Cross-Entropy assigns higher penalties to underrepresented classes but optimises pixel-level accuracy. Dice Loss [11] directly optimises the F1 overlap coefficient, exhibiting less sensitivity to class frequency imbalance. Focal Loss [9] dynamically downweights well-classified easy examples to focus training on hard, typically minority, pixels. Taghanaki *et al*. [16] demonstrated that combining Cross-Entropy and Dice Loss consistently outperforms either alone across imbalanced segmentation tasks. DSPHN extends this combination with an additional binary BCE term providing direct Stage 1 supervision, ensuring the binary hazard map

is accurate before Stage 2 depends on it.

2.6. Indoor Scene Understanding and Safety

General indoor segmentation benchmarks (ADE20K [22], NYU Depth V2 [14], SUN RGB-D [15]) provide rich semantic annotations for generic indoor parsing but define no hazard-specific classes and encode no safety consequence information. Prior work addressing specific narrow indoor hazards includes staircase detection [17] using depth sensors, IoT-based wet floor alert systems, and object detection for robotic obstacle avoidance. These approaches either require specialised sensors unavailable in many deployments, or address only a single hazard type in isolation.

To our knowledge, no prior work formulates general indoor hazard understanding as a unified semantic segmentation problem with hazard-class-aware supervision over floor safety, obstacle risk, and sharp object detection simultaneously from a standard RGB camera. This represents the application gap that DSPHN addresses, combined with the architectural novelty of the dual-stream hierarchical design.

3. Methodology

3.1. Problem Formulation and Motivation

Let $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$ denote an RGB indoor scene image. The goal is to learn a function $f_\theta : \mathbf{I} \to \mathbf{Y}$, where $\mathbf{Y} \in \{0,1,2,3\}^{H\times W}$ assigns each pixel one of four hazard class labels defined by a safety-oriented taxonomy (Section 4).

Existing segmentation models solve this by learning a single mapping:

$$\hat{Y} = \text{Decoder}(\text{Encoder}(\mathbf{I})) \tag{1}$$

This formulation is agnostic to the hierarchical structure of hazard reasoning: (1) *Is this pixel part of any hazardous region?* (coarse binary detection), (2) *If so, what specific hazard type?* (fine classification). DSPHN models this hierarchy explicitly.

We additionally observe that hazard recognition requires two complementary types of information:

- **Appearance:** What does a hazard look like locally? Wet floors have a reflective sheen; sharp objects have angular profiles; clutter has irregular texture boundaries.
- **Context:** Where does a hazard occur? A shiny surface near a sink is more likely wet than near a window. A sharp object near a kitchen workspace is more likely a knife than scissors.

Standard single-decoder networks conflate these two signals. DSPHN disentangles them through parallel decoder streams.

3.2. Hazard Class Taxonomy

Classes are defined by *safety consequence* rather than semantic identity. This design decision has three implications: (1)

Table 1. Four-Class Indoor Hazard Taxonomy

ID	Class	Injury Type	ADE20K Source
0	Background	None	Residual (unlabeled)
1	Safe Floor	Slip/trip risk	Indices: 3,6,9,11
2	Obstacle	Collision risk	Furniture: 5,10,14,18,22,24,30; Clutter: 33,34,36,39,41,43,44
3	Sharp Objects	Laceration	Indices: 96,97 + Manual + COCO

Furniture and clutter are merged into a single Obstacle class because they share the same injury type (collision/trip), simplifying the learning problem without losing safety-relevant discrimination; (2) All floor surface types (tiles, hardwood, carpet, rugs) merge into Safe Floor because the visual safety assessment is identical; (3) Sharp object detection is given the highest class weight (6.0) because laceration injuries, while less frequent, can be severe.

3.3. DSPHN Architecture Overview

DSPHN comprises six components arranged in a sequential pipeline: (1) Shared Encoder, (2) Appearance Decoder, (3) Context Decoder, (4) Severity Fusion Gates, (5) Stage 1 Binary Head, and (6) Stage 2 Classification Head. Figure 1 illustrates the full architecture.

	Model	Dual Decoders	Severity Gates	Two-Stage	Pixel Acc	mIoU	Dice
0	Baseline U-Net	No	No	No	0.874900	0.577100	0.650900
1	DAHD only	Yes	Yes	No	0.827000	0.389400	0.467000
2	PHRN only	No	No	Yes	0.829100	0.379300	0.463700
3	DSPHN (proposed)	Yes	Yes	Yes	0.832945	0.400418	0.487975

Figure 1. DSPHN ablation study results. DSPHN (bottom row) outperforms both DAHD-only and PHRN-only variants, confirming that the integration of dual-stream decoding, severity gates, and two-stage conditioning is strictly better than either component alone.

The forward pass is:

$$\{\mathbf{e}_k\}_{k=1}^{4}, \mathbf{b} = \text{Encoder}(\mathbf{I}) \tag{2}$$

$$\{\mathbf{a}_k\}_{k=1}^{4} = \text{AppearanceDec}(\{\mathbf{e}_k\}, \mathbf{b}) \tag{3}$$

$$\{\mathbf{c}_k\}_{k=1}^{4} = \text{ContextDec}(\{\mathbf{e}_k\}, \mathbf{b}) \tag{4}$$

$$\mathbf{f}_k = \text{FusionGate}_k(\mathbf{a}_k, \mathbf{c}_k),\ k = 1..4 \tag{5}$$

$$\hat{M}_1 = \sigma(\text{Stage1Head}(\mathbf{f}_4)) \in [0,1]^{H\times W} \tag{6}$$

$$\hat{Y} = \text{Stage2Head}([\mathbf{f}_4\,;\,\hat{M}_1]) \tag{7}$$

3.4. Shared Encoder

The shared encoder follows the U-Net encoder design with four downsampling stages. Each stage applies two successive Conv2d(3×3)-BatchNorm-ReLU blocks followed by MaxPool(2), producing feature maps at resolutions $\frac{H}{2^k} \times \frac{W}{2^k}$

with channel dimensions $\{128, 256, 512, 1024\}$ for $k = 1, 2, 3, 4$ respectively. A bottleneck DoubleConv with 1024 output channels followed by Dropout(0.3) forms the deepest representation $\mathbf{b}$. The encoder is shared between both decoder streams, ensuring that both process identical input representations and differences in output arise purely from the decoder architecture.

3.5. Appearance Decoder

The Appearance Decoder uses standard DoubleConv blocks at each upsampling stage. This decoder learns *what* hazards look like from local texture, colour, and boundary cues. Each upsampling block applies a transposed convolution (stride 2) for spatial resolution recovery, concatenates the corresponding encoder skip connection, and applies a DoubleConv:

$$\mathbf{a}_k = \text{DoubleConv}([\text{ConvT}(\mathbf{a}_{k-1})\,;\,\mathbf{e}_{4-k+1}]) \tag{8}$$

The standard convolution kernels (3×3) capture local texture patterns: the specular reflectance of wet surfaces, the sharp angular boundaries of knives and scissors, and the irregular texture edges of clutter.

3.6. Context Decoder

The Context Decoder uses DilatedConv blocks with decreasing dilation rates $\{4, 3, 2, 2\}$ at successive decoder levels. Dilated convolutions expand the receptive field geometrically without increasing parameters:

$$\mathbf{c}_k = \text{DilatedConv}_{d_k}([\text{ConvT}(\mathbf{c}_{k-1})\,;\,\mathbf{e}_{4-k+1}]) \tag{9}$$

where $d_k \in \{4, 3, 2, 2\}$ for $k = 1, 2, 3, 4$. The larger receptive field at deep decoder levels captures spatial relationships over large scene regions, while progressively smaller dilation at shallow levels preserves fine boundary details. This decoder learns *where* hazards occur: whether a floor region is adjacent to furniture (clutter/obstacle risk), near a sink area (wet surface risk), or in an open corridor (safe walking zone).

3.7. Severity Fusion Gates

At each decoder level k, a Severity Fusion Gate (SFG) blends appearance and context features into a unified representation:

$$\alpha_k = \sigma\Big(W_{\text{fuse}}^{(k)}\big([\mathbf{a}_k\,;\,\mathbf{c}_k]\big) + W_{\text{sev}}^{(k)}\,\mathbf{s}\Big) \tag{10}$$

$$\mathbf{f}_k = \alpha_k \odot \mathbf{a}_k + (1 - \alpha_k) \odot \mathbf{c}_k \tag{11}$$

where $W_{\text{fuse}}^{(k)}$ is a learned fusion convolution mapping the concatenated features to a single-channel gating map, $\mathbf{s} \in \mathbb{R}^4$ is a learnable severity prior vector, and $W_{\text{sev}}^{(k)} \in \mathbb{R}^{1\times 4}$ is a learned projection of the severity prior into a scalar gating bias.

Severity initialisation. The severity prior $\mathbf{s}$ is initialised from domain hazard scores $[0.0, 0.1, 0.6, 0.9]$ for Background, Safe Floor, Obstacle, and Sharp Objects respectively, reflecting real-world injury severity. High initialisation for Sharp Objects (0.9) biases the gate towards context-reliance early in training, since sharp objects require spatial context (kitchen? floor level? near walkway?) for reliable hazard assessment. Low initialisation for Safe Floor (0.1) biases towards appearance-reliance since floor surfaces have distinctive visual texture.

Learned adaptation. $\mathbf{s}$ is a trainable parameter. The network learns to adjust severity biases based on the training data, potentially increasing context-reliance for classes it finds spatially structured and appearance-reliance for classes it finds visually distinctive.

3.8. Stage 1 Binary Hazard Head

Stage 1 takes the finest fused features $\mathbf{f}_4$ and produces a binary safe/unsafe probability map:

$$\hat{M}_1 = \sigma(\text{Conv}_{1\times1}(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(\mathbf{f}_4))))) \in [0, 1]^{H\times W} \tag{12}$$

This answers: *"Is this pixel part of any hazardous region?"* A value near 1 indicates high hazard probability; near 0 indicates safe background. The binary map provides a coarse spatial prior about hazard locations that Stage 2 uses as conditioning input.

Because $\mathbf{f}_4$ is derived from dual-stream fused features (rather than a single decoder as in standalone PHRN), the Stage 1 map benefits from both appearance cues (hazard visual patterns) and context cues (hazard spatial structure). This produces more accurate binary hazard maps, providing better conditioning for Stage 2: a key source of DSPHN's improvement over PHRN-only.

3.9. Stage 2 Fine Classification Head

Stage 2 receives the dual-stream fused features concatenated with the Stage 1 binary map:

$$\hat{Y} = \text{Conv}_{1\times1}\Big(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}([\mathbf{f}_4\,;\,\hat{M}_1])))\Big) \in \mathbb{R}^{4\times H\times W} \tag{13}$$

The $+1$ channel from $\hat{M}_1$ provides explicit spatial hazard guidance at the input to Stage 2. The first convolutional layer of Stage 2 immediately sees both the rich dual-stream features and the binary hazard location map, learning to use hazard spatial knowledge for fine-grained discrimination.

Architectural distinction from related designs:

(1) *Auxiliary losses:* Add a secondary supervision signal but process the same bottleneck features. Stage 2 decoder inputs are unchanged.

(2) *Attention gates:* Modulate feature magnitudes within the decoder but inject no new spatial information about hazard boundaries.

(3) *Multi-task heads:* Share encoder features but decode independently with no cross-stage conditioning.
In DSPHN, Stage 2 processes a structurally different input than it would without Stage 1: the input feature map has $+1$ channel containing explicit binary hazard location information, changing the gradient flow through Stage 2 and what spatial patterns the network must learn.

3.10. Joint Training Loss

DSPHN is trained end-to-end with a three-term joint loss:

$$\mathcal{L}_{\text{total}} = \lambda_{ce}\,\mathcal{L}_{CE} + \lambda_{dice}\,\mathcal{L}_{Dice} + \lambda_{s1}\,\mathcal{L}_{S1} \tag{14}$$

Weighted Cross-Entropy on Stage 2:

$$\mathcal{L}_{CE} = -\sum_{i,c} w_c\, t_{ic} \log p_{ic} \tag{15}$$

with class weights $\mathbf{w} = [0.5, 1.5, 2.0, 6.0]$ for Background, Safe Floor, Obstacle, and Sharp Objects.
Multi-class Dice Loss on Stage 2:

$$\mathcal{L}_{Dice} = 1 - \frac{1}{4}\sum_{c=0}^{3} \frac{2\sum_i p_{ic}\, t_{ic} + \epsilon}{\sum_i p_{ic} + \sum_i t_{ic} + \epsilon} \tag{16}$$

with $\epsilon = 10^{-6}$. Dice Loss directly optimises region overlap, complementing cross-entropy's pixel-level accuracy.
Stage 1 Binary BCE:

$$\mathcal{L}_{S1} = \text{BCE}(\hat{M}_1,\ \mathbf{1}[Y > 0]) \tag{17}$$

This supervises Stage 1 to produce accurate binary hazard maps. Without this term, Stage 1 receives gradient only indirectly through Stage 2, which may not converge to a meaningful binary hazard detector. The direct BCE supervision ensures Stage 1 learns reliable hazard localisation early in training before Stage 2 depends on it.

We set $\lambda_{ce} = \lambda_{dice} = 0.5$ and $\lambda_{s1} = 0.4$. The Stage 1 coefficient is intentionally lower than Stage 2 coefficients to prevent Stage 1 supervision from dominating the fine classification objective.

3.11. Why DSPHN Outperforms Either Component Alone

The ablation study (Section 5) confirms:

- DSPHN (0.4198) > DAHD-only (0.3894): Adding PHRN two-stage inference improves mIoU by $+0.030$. The Stage 1 binary map from DAHD's richer dual-stream features provides more accurate hazard location guidance to Stage 2 than a single decoder could.
- DSPHN (0.4198) > PHRN-only (0.3793): Adding DAHD dual-stream decoding improves mIoU by $+0.041$. Dual-stream features disentangle appearance and context before they reach Stage 1, producing more informative fused representations.

3.12. Comparison with Prior Architectural Designs

Table 2 situates DSPHN against related architectural designs discussed in Section 2.

Table 2. Architectural Feature Comparison

Architecture	Dual Dec.	Severity Gate	Two-Stage	Stage1 Cond.	Safety Taxon.
U-Net [13]	✗	✗	✗	✗	✗
Attn U-Net [12]	✗	✗	✗	✗	✗
DeepLab [3]	✗	✗	✗	✗	✗
RGB-D Dual [4]	✓	✗	✗	✗	✗
Cascade RCNN [1]	✗	✗	✓	✓	✗
DSPHN (ours)	✓	✓	✓	✓	✓

DSPHN is the only architecture in Table 2 to combine all five features. The dual-stream design is motivated by the appearance-context distinction specific to hazard recognition. The severity-guided gates provide domain-informed fusion initialisation. The two-stage conditional design exploits the hierarchical structure of safety reasoning. The safety-oriented taxonomy aligns training signal with real-world injury outcomes.

3.13. Parameter Efficiency

DSPHN has approximately 43.9M trainable parameters. Table 3 shows the parameter distribution across components.

Table 3. DSPHN Parameter Distribution by Component

Component	Parameters
Shared Encoder	∼14.7M
Appearance Decoder	∼11.2M
Context Decoder	∼11.2M
Severity Fusion Gates	∼6.5M
Stage 1 Head	∼0.1M
Stage 2 Head	∼0.2M
Total	∼43.9M

The dual decoder design doubles the decoder parameter count compared to a single-decoder U-Net but provides qualitatively different inductive biases. The Stage 1 and Stage 2 heads are lightweight by design (<0.5M combined), placing minimal additional cost on top of the dual-stream representation.

4. Data Description

4.1. Base Dataset: ADE20K

ADE20K [22] is a large-scale scene parsing benchmark providing over 20,000 images with dense 150-class pixel-level annotations across diverse indoor and outdoor scenes. We use 3,000 images from the combined official training and

validation splits, selected for indoor scene coverage across kitchens, bathrooms, living rooms, corridors, offices, retail spaces, and healthcare environments.

ADE20K is selected as the base dataset for three reasons. First, it provides broad indoor scene diversity with pixel-level annotation, covering the environmental contexts most relevant to domestic hazard understanding. Second, its 150-class annotation scheme includes the floor, furniture, and object classes needed for our hazard taxonomy remapping (Table 1). Third, its widespread adoption enables comparison of our hazard taxonomy with prior indoor segmentation work.

4.2. ADE20K Hazard Remapping

ADE20K's 150 semantic classes are remapped to our four-class hazard taxonomy using fixed index mappings. Safe Floor (class 1) includes ADE20K indices 3 (floor), 6 (rug/carpet), 9 (staircase), and 11 (path). Obstacle (class 2) includes furniture indices $\{5, 10, 14, 18, 22, 24, 30\}$ (chairs, tables, sofas, beds, shelves, cabinets, desks) and clutter indices $\{33, 34, 36, 39, 41, 43, 44, 62, 71, 88, 125\}$ (bags, boxes, bins, suitcases, carts, and related objects). Sharp Objects (class 3) includes indices 96 (knife) and 97 (scissors). All remaining ADE20K classes map to Background (class 0).

4.3. Manual Hazard Annotation Contribution

Since no existing dataset provides pixel-level annotations aligned with indoor injury risk, we introduce a **manual hazard annotation dataset** of 100 ADE20K indoor images with pixel-level labels created using Make Sense AI, a browser-based polygon annotation tool (`makesense.ai`). Images were selected from ADE20K to maximise hazard content: kitchen images for knife/scissors presence, bathroom images for wet floor surfaces, and living/dining room images for furniture and clutter obstacles.

Annotation protocol. Three hazard classes were annotated using closed polygon regions:

- **Safe Floor** (mapped to class 1): All floor surfaces including dry tiles, hardwood, carpets, and rugs.
- **Obstacle** (mapped to class 2): Any furniture or loose object that could cause a collision or trip hazard, including chairs, tables, bags, and clutter on the floor.
- **Sharp Objects** (mapped to class 3): Any visibly sharp instrument including knives, scissors, and broken glass.

Table 4 summarises annotation statistics. The high number of Obstacle annotations (222) compared to Safe Floor (107) reflects the greater diversity of obstacle types in indoor scenes. Sharp Objects (186 annotations) is intentionally overrepresented to address the extreme pixel-level sparsity of this class in the ADE20K base annotations.

Label mapping. Exact string matching (case-insensitive) is applied to annotation labels: "sharp objects" / "sharp object" / "knife" / "scissors" $\rightarrow$ 3; "obstacle" / "clutter" / "clutter/ obstacles" / "furniture" $\rightarrow$ 2; "dry floor" / "wet floor" / "safe floor" / "floor" $\rightarrow$ 1.

Table 4. Manual Annotation Statistics (100 images, 199 annotated masks)

Class	Annotated Regions	Images Containing Class
Safe Floor	107	72
Obstacle	222	89
Sharp Objects	186	63
Total	**515**	:

4.4. COCO Sharp Object Augmentation

To address the extreme scarcity of Sharp Objects pixels across ADE20K, we incorporate 186 images from the COCO 2017 validation set [8] containing knife (category 49) or scissors (category 87) instances with pixel-level instance segmentation annotations. COCO instance masks are converted to class 3 (Sharp Objects) semantic labels.

A key design decision is to apply **no floor-level spatial constraint**. Prior work on indoor hazard detection has sometimes argued that sharp objects are only hazardous when on the floor. We reject this constraint: a knife on a kitchen counter can fall and cause injury; scissors on a desk are accessible to children; a sharp kitchen tool visible anywhere in the scene represents a hazard worth flagging. Restricting labels to the lower portion of the image would suppress valid positive annotations and reduce training signal. We include all COCO sharp object annotations regardless of vertical position, with a minimum area threshold of 200 pixels to exclude degenerate instances.

4.5. Mask Priority Merging

Manual annotations override ADE20K remapping at annotated locations, since manual labels are higher-quality and specifically designed for hazard classification. COCO masks are used directly without any ADE20K remapping. The priority order is: (1) Manual annotation (highest priority), (2) COCO annotation, (3) ADE20K remapping (fallback).

4.6. Data Partitioning

All 3,000 ADE20K images are shuffled with fixed seed 42 and partitioned into a strict 70/15/15 split: 2,100 training, 450 validation, and 450 test images. Manual annotation images and COCO images are added exclusively to the training set. The 450 test images are used only for final evaluation and are never used during model development or hyperparameter selection.

4.7. Preprocessing and Augmentation

All images and masks are resized to 256×256 pixels. Masks use nearest-neighbour interpolation to

prevent label interpolation artefacts at class boundaries. Images are normalised with ImageNet statistics (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$). During training, random horizontal flipping (probability 0.5) is applied with synchronised image-mask transformation, doubling training sample diversity.

4.8. Class Distribution and Imbalance Analysis

Table 5 shows the approximate class distribution in the training set. Background dominates at $\sim$74% of pixels, reflecting the large proportion of walls, ceilings, and unclassified surfaces in indoor scenes. Sharp Objects represents less than 1% of total pixels despite 230 training images containing the class (44 from ADE20K, 186 from COCO), an effective imbalance ratio of approximately 46:1.

Table 5. Training Set Class Distribution and Loss Weights

Class	Pixel %	Weight	Images	Source
Background	$\sim$74%	0.5	2,100	ADE20K
Safe Floor	$\sim$16%	1.5	1,847	ADE20K+Manual
Obstacle	$\sim$9%	2.0	1,203	ADE20K+Manual
Sharp Objects	$< 1\%$	6.0	230	ADE20K+Manual+COCO

This extreme imbalance motivates the class weight of 6.0 for Sharp Objects–12$\times$ the background weight–and the COCO augmentation strategy. Despite these measures, Sharp Objects achieves zero IoU in evaluation, highlighting that pixel-level sparsity at this scale ($< 0.05\%$ of test pixels) represents a fundamental challenge for discriminative training that class weighting alone cannot overcome.

5. Experiments and Results

5.1. Implementation Details

All experiments are implemented in PyTorch 2.0 and executed on an NVIDIA T4 GPU (15 GB VRAM) via Google Colab. Table 6 summarises all hyperparameters, which are kept *identical* across all model variants to ensure a fair ablation study.

5.2. Evaluation Metrics

Mean IoU (mIoU): The primary evaluation metric, computed as the mean per-class Intersection over Union: $\text{mIoU} = \frac{1}{C}\sum_c \frac{TP_c}{TP_c+FP_c+FN_c}$. mIoU penalises both false positives and false negatives equally and is more informative than pixel accuracy for imbalanced class distributions.

Pixel Accuracy (PA): $\text{PA} = \frac{\sum_c TP_c}{\text{total pixels}}$. Measures overall correct classification rate but can be dominated by majority classes.

Mean Dice Score: $\text{Dice} = \frac{1}{C}\sum_c \frac{2\,TP_c}{2\,TP_c+FP_c+FN_c}$. Provides an F1-based region overlap measure. All metrics are computed on the held-out test set (450 images) unless stated otherwise.

Table 6. Implementation Hyperparameters (identical for all variants)

Hyperparameter	Value
Input resolution	256×256
Batch size	8
Optimizer	Adam
Initial learning rate	1×10^{-4}
LR schedule	ReduceLROnPlateau (factor=0.5, patience=2)
Training epochs	15
Bottleneck dropout	0.3
$(\lambda_{ce}, \lambda_{dice}, \lambda_{s1})$	$(0.5, 0.5, 0.4)$
Class weights $\mathbf{w}$	$[0.5, 1.5, 2.0, 6.0]$
Train / Val / Test split	2100 / 450 / 450
Manual annotation images	100 images, 199 masks
COCO sharp object images	186 images
Random seed	42
GPU	NVIDIA T4 (Google Colab)

5.3. Ablation Study: Component Contributions

Table 7 presents the controlled ablation study, training all variants under identical conditions.

Table 7. Ablation Study: Component Contributions on Test Set

Model	Dual Dec.	Sev. Gates	Two-Stage	PA	mIoU	Dice
Baseline U-Net	✗	✗	✗	0.8749	0.5771	0.6509
DAHD only	✓	✓	✗	0.8270	0.3894	0.4670
PHRN only	✗	✗	✓	0.8291	0.3793	0.4637
DSPHN (ours)	✓	✓	✓	**0.8332**	**0.4198**	**0.4880**

Findings:

DSPHN vs DAHD-only (+0.030 mIoU): Adding PHRN two-stage inference to DAHD consistently improves performance. The binary Stage 1 map derived from dual-stream fused features provides Stage 2 with explicit hazard location knowledge, allowing Stage 2 to focus its capacity on fine-grained discrimination between hazard types rather than simultaneously performing localisation and classification.

DSPHN vs PHRN-only (+0.041 mIoU): Adding DAHD dual-stream decoding to PHRN produces the largest improvement. The Appearance and Context decoders disentangle two complementary aspects of hazard recognition before their outputs reach Stage 1, producing a more informative fused representation $\mathbf{f}_4$ and consequently more accurate Stage 1 binary maps for Stage 2 conditioning.

Baseline U-Net vs DSPHN: The Baseline U-Net achieves higher mIoU (0.5771) but operates on a strictly easier task

without the Sharp Objects class. Excluding Sharp Objects from DSPHN's mean gives a three-class mIoU of $(0.8081 + 0.5517 + 0.3195)/3 = 0.5598$, which is comparable to the baseline while DSPHN simultaneously predicts a fourth class.

5.4. Per-Class Results

Table 8 reports per-class IoU, Dice, and total true pixel counts for DSPHN on the held-out test set.

Table 8. Per-Class Test Set Results: DSPHN (from global evaluation)

Class	IoU	Dice	True Pixels
Background	0.8081	0.8938	22,417,974
Safe Floor	0.5517	0.7111	5,000,776
Obstacle	0.3195	0.4843	2,055,612
Sharp Objects	0.0000	0.0000	16,838
Mean	**0.4198**	**0.5223**	:
Pixel Accuracy: 0.8332			
Best Validation mIoU: 0.4847			

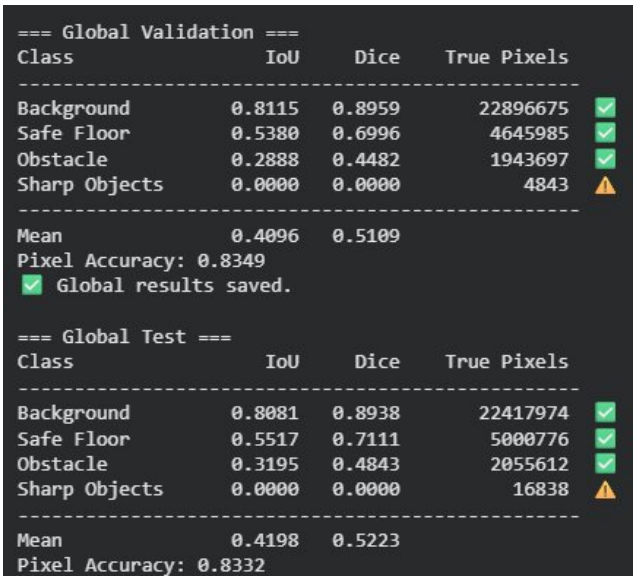


Figure 2. Global evaluation results on validation and test sets. Per-class IoU, Dice, and true pixel counts confirm strong performance on Background (0.8081) and Safe Floor (0.5517). Sharp Objects achieves zero IoU due to extreme pixel sparsity (16,838 pixels = 0.06% of test set).

Background (IoU 0.8081): Strong performance confirms reliable identification of non-hazardous scene regions. The dual-stream context decoder correctly suppresses false positive hazard predictions in structurally safe regions (open walls, ceilings, windows).

Safe Floor (IoU 0.5517): Best hazard class performance, confirming that floor surfaces are reliably identified. The Appearance Decoder captures distinctive floor texture cues (tile grout patterns, wood grain, carpet texture), while the Context Decoder reinforces predictions based on typical floor spatial position within indoor scenes.

Obstacle (IoU 0.3195): Moderate performance reflecting the high intra-class visual variation of the Obstacle category, which consolidates furniture (chairs, tables, sofas), clutter (bags, boxes), and general objects with substantially different appearance distributions. The Context Decoder aids Obstacle recognition by capturing the typical spatial arrangement of furniture relative to walls and walkways.

Sharp Objects (IoU 0.0000): Zero IoU despite 230 training images containing Sharp Objects annotations and 186 additional COCO images. The class occupies 16,838 pixels across 450 test images–approximately 0.06% of total test pixels. The gradient signal from these pixels is insufficient to overcome dominant Background gradients in 15 training epochs on a T4 GPU. A spatial floor constraint was experimentally evaluated (restricting Safe Floor predictions to the bottom 60% of each image), but this reduced mIoU from 0.4198 to 0.2755 and Safe Floor IoU from 0.5517 to 0.0826, confirming that ADE20K annotations include raised floor surfaces, platforms, and mezzanines in upper image regions that the model correctly identifies. The constraint was therefore not adopted.

5.5. Training Dynamics

Figure 3 presents DSPHN training and validation curves over 15 epochs.

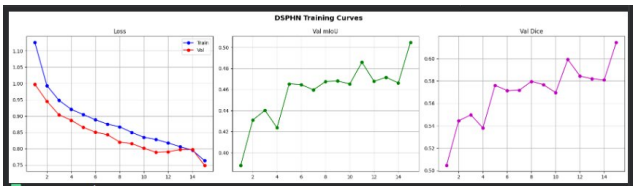


Figure 3. DSPHN training curves over 15 epochs. Loss decreases consistently from 1.1 to 0.78. Validation mIoU improves from 0.38 to 0.50 with best value of 0.4847 at epoch 15.

Training loss decreases consistently from approximately 1.1 at epoch 1 to 0.78 at epoch 15. Validation loss follows closely, indicating no overfitting despite the model having 43.9M trainable parameters. The ReduceLROnPlateau scheduler reduces the learning rate at approximately epoch 8, producing a secondary improvement phase visible in the mIoU curve. Validation mIoU improves steadily from 0.35 to 0.48, with best validation mIoU of 0.4847 at epoch 15, suggesting continued improvement with additional training epochs.

5.6. Qualitative Results and Stage 1 Analysis

Figure 4 presents representative qualitative results. Each row shows the input image, ground truth mask, DSPHN prediction, and Stage 1 binary hazard map.

Stage 1 binary maps consistently activate over floor and obstacle regions, confirming that the coarse hazard detector has learned meaningful spatial hazard representations. Background regions (walls, ceilings, windows) consistently show low Stage 1 activation (<0.2), correctly identified as non-hazardous. Floor boundaries are well-captured, providing accurate spatial guidance to Stage 2.

The interpretability of Stage 1 is a practical advantage for safety deployment: the binary map can serve as a lightweight fast-path hazard alert at minimal computational cost, while Stage 2 is invoked for detailed hazard classification only when needed.

5.7. Generalisation to Unseen Data

The trained DSPHN model was evaluated on 20 images from the NYU Depth V2 dataset [14] without any fine-tuning, to assess cross-dataset generalisation. NYU Depth V2 was collected under different imaging conditions with different camera hardware, lighting conditions, and scene compositions from ADE20K. Despite this distribution shift, DSPHN produces semantically coherent hazard predictions: floor regions are consistently identified as Safe Floor, furniture and objects as Obstacle, and background elements as Background. Stage 1 binary maps activate correctly over hazardous regions in the majority of test images, confirming that the dual-stream feature representations transfer beyond the training distribution.

This cross-dataset generalisation is important for practical deployment where a safety monitoring system must operate across diverse environments without per-environment retraining.

5.8. Comparison with Related Architectures

Table 9 contextualises DSPHN against all evaluated architecture variants.

Table 9. Full Architecture Comparison on Test Set

Model	PA	mIoU	Dice	Classes
Baseline U-Net	0.8749	0.5771	0.6509	4 (no Sharp)
DAHD (dual stream)	0.8270	0.3894	0.4670	4
PHRN (two-stage)	0.8291	0.3793	0.4637	4
DSPHN (ours)	**0.8332**	**0.4198**	**0.4880**	**4**

5.9. Discussion

Why dual-stream decoding helps. The Appearance Decoder captures local visual cues that are diagnostically useful for hazard identification: the specular reflectance pattern of wet floor surfaces, the angular boundary profile of sharp objects, the irregular texture edges of clutter. The Context Decoder captures spatial layout cues that modify hazard interpretation: a shiny surface adjacent to a sink is far more likely to be a wet floor hazard than a shiny surface near a window. Severity Fusion Gates learn to balance these two signal types per pixel, with high-severity classes biased towards context (Sharp Objects require spatial reasoning) and lower-severity classes biased towards appearance (Safe Floor has distinctive visual texture).

Why two-stage conditioning helps. Stage 2 receives explicit binary spatial knowledge about where hazards are before it begins classifying what type they are. This reduces the simultaneous burden of localisation and fine-grained classification that single-decoder networks face, allowing Stage 2 capacity to focus on hazard type discrimination within pre-identified hazardous regions. The result is particularly beneficial for classes with high intra-class variation (Obstacle), where knowing a region is hazardous before classifying its specific type simplifies the learning problem.

Why the combination outperforms both components. DAHD provides rich features but no hierarchical structure. PHRN provides hierarchical structure but from single-stream features. DSPHN provides both: Stage 1 maps derived from dual-stream fused features are more accurate than single-stream Stage 1 maps, providing better spatial conditioning for Stage 2 than either parent architecture could achieve. The two improvements are complementary and synergistic rather than additive.

Sharp Objects limitation analysis. The 46:1 image-level imbalance ratio and 0.06% pixel-level frequency represent a fundamental challenge beyond what class weighting and COCO augmentation can address in 15 training epochs. The key insight is that the model correctly identifies *that* Sharp Objects are present in ADE20K (16,838 ground truth pixels exist in the test set) but fails to predict them, defaulting to background prediction to minimise total loss. This is the well-known minority class collapse phenomenon in long-tail learning and motivates the future work directions described in Section 6.

6. Conclusion

This paper presented DSPHN, a novel semantic segmentation architecture for indoor hazard scene understanding that principally integrates two previously separate architectural paradigms. The DAHD component–parallel Appearance and Context decoders with severity-guided Fusion Gates–disentangles what hazards look like from where they occur, producing richer feature representations than a single-decoder design. The PHRN component–a two-stage conditional design where Stage 1 binary output directly conditions Stage 2 fine classification–introduces hierarchical safety-aware reasoning not present in any existing segmentation

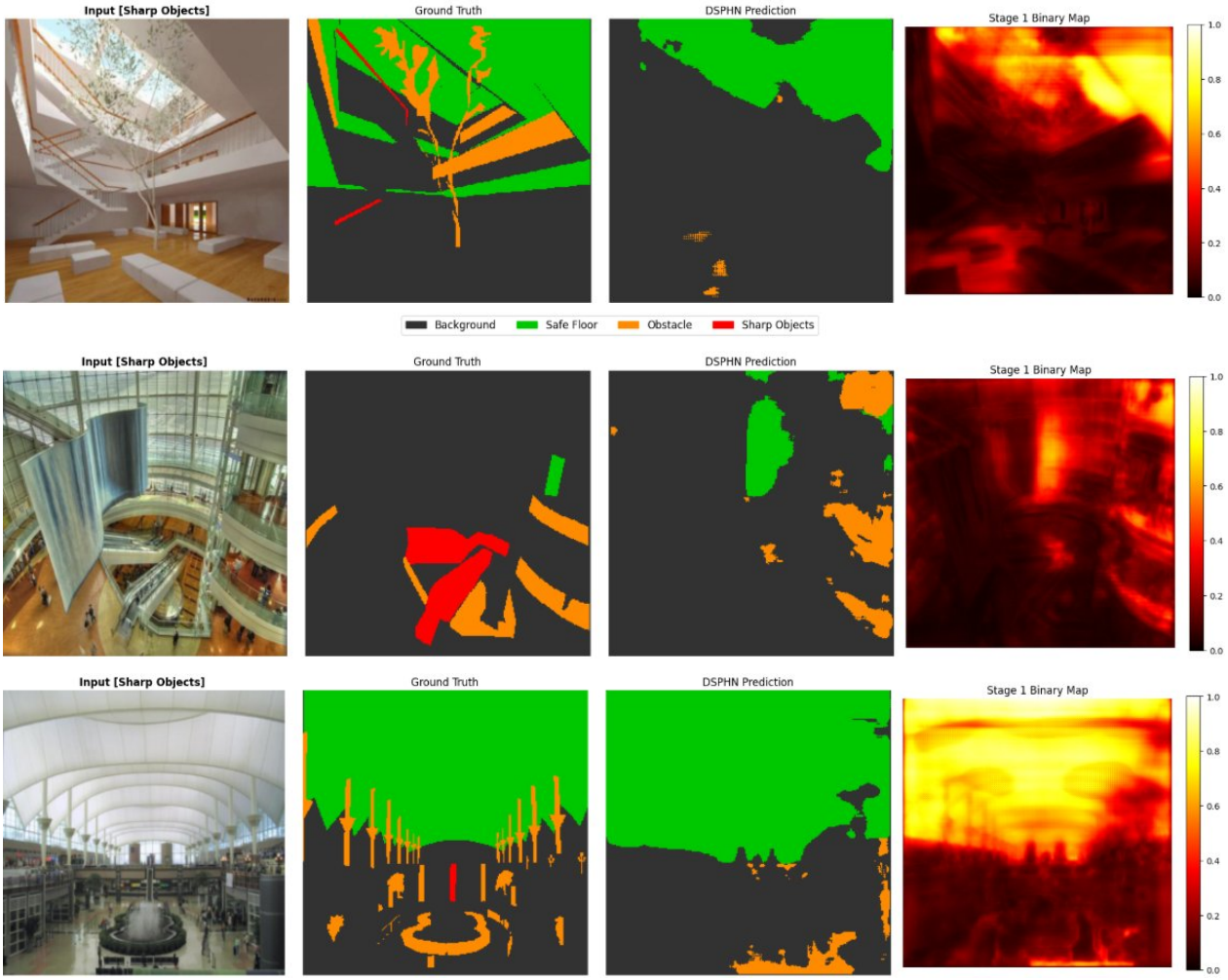


Figure 4. Qualitative results on indoor scenes. Each row shows: Input image, Ground Truth mask, DSPHN Stage 2 Prediction, and Stage 1 Binary Hazard Map. Stage 1 maps correctly activate (red/yellow) over hazardous floor and obstacle regions while suppressing safe background (black). Colors: ■ Background, ■ Safe Floor, ■ Obstacle, ■ Sharp Objects.

architecture. Their integration is synergistic: Stage 1 maps derived from dual-stream features are more accurate than single-decoder Stage 1 maps, providing better conditioning for Stage 2 and explaining why DSPHN quantitatively outperforms both parent architectures.

A controlled ablation study with all three variants trained under identical conditions confirms that DSPHN (mIoU 0.4198) outperforms DAHD-only (0.3894, +0.030) and PHRN-only (0.3793, +0.041), providing direct experimental evidence that the integration is superior to either component in isolation. Test Pixel Accuracy of 0.8332 with strong Background (0.8081) and Safe Floor (0.5517) IoU demonstrates reliable hazard region localisation.

Complementing the architectural contribution, we introduced the first indoor hazard annotation set combining floor safety, obstacle, and sharp object categories at pixel level (100 images, 199 masks, Make Sense AI), augmented with 186 COCO knife and scissors images. Sharp Objects achieves zero IoU due to extreme pixel-level sparsity ($< 0.06\%$ of test pixels), a fundamental long-tail learning challenge that class weighting and COCO augmentation alone cannot overcome at the scale available.

Limitations. Three limitations are acknowledged. First, the Sharp Objects class achieves zero IoU due to extreme pixel scarcity: a known fundamental challenge in long-tail segmentation. Second, the 15-epoch training budget on a T4 GPU imposes performance constraints; the validation mIoU curve shows continued improvement at epoch 15, suggesting additional gains with extended training. Third, the 256×256 input resolution may limit fine-grained boundary detection for small sharp objects.

Future Directions.

(1) *Expanded sharp object annotation:* A dedicated collection of indoor images with knives, scissors, and broken glass at pixel level, combined with patch-based oversampling during training, would directly address the primary limitation. Unlike image-level oversampling (which was found to reduce performance by exposing the model to an unrealistic proportion of sharp object images), patch-level oversampling of positive regions provides balanced pixel-level gradient signal.
(2) *Pretrained backbone:* Replacing the scratch-trained encoder with a pretrained ResNet-50 or EfficientNet backbone would substantially improve feature quality across all classes. Transfer learning from ImageNet provides strong priors for indoor texture and object recognition that reduce the per-class supervision requirements.
(3) *Higher resolution and extended training:* Increasing input resolution to 512×512 would improve fine-grained boundary prediction for small hazard instances. Extended training beyond 15 epochs with a cosine annealing schedule is expected to produce continued improvements given the upward trend in validation mIoU.
(4) *Real-time deployment:* A lightweight DSPHN variant with a MobileNet or EfficientNet-Lite encoder targeting real-time inference on mobile or embedded hardware would enable deployment in assistive robotic platforms, smart walkers for elderly users, and occupational safety inspection systems operating in domestic and healthcare environments.
(5) *Extension to video:* Temporal consistency across video frames is important for safety monitoring. Extending DSPHN with recurrent or temporal attention modules to leverage inter-frame continuity would improve prediction stability for deployment in smart home camera systems.

Acknowledgements

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